

A disconnected nonzero fixed-diagonal idempotent fiber in infinite-dimensional Hilbert space

Candidate counterexample for arXiv:0906.0139

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Status: candidate counterexample, likely valid; human review pending. The construction is self-contained and works over both real and complex scalars.

Source question

Giol, Kovalev, Larson, Nguyen, and Tener prove that for every diagonal $d \in D_n(\mathbb{C})$, the finite-dimensional fiber

$$\{q \in M_n(\mathbb{C}) : q^2 = q, E(q) = d\}$$

is path-connected. At the end of their introduction they ask whether full analogues of this theorem and their half-diagonal projection theorem hold on separable infinite-dimensional Hilbert spaces, in the operator-norm topology [1]. The relevant source context spans PDF pages 2–3:

scalars Theorem 1.1 remains true if $n \geq 2$, see Remark 3.2. Therefore, $\mathcal{F}_{2n,n}^{\mathbb{R}}$ has at most two connected components when $n \geq 2$, and its quotient under the natural action of the orthogonal group in $\mathbb{R}^n$ is connected.

We denote by $M_n(\mathbb{C})$ (resp. $M_n(\mathbb{R})$) the set of all $n \times n$ matrices with complex (resp. real) entries. When the choice of $\mathbb{C}$ or $\mathbb{R}$ is unimportant, we write simply M_n . Let $D_n \subset M_n$ be the subalgebra of diagonal matrices. There is a natural linear operator (conditional expectation) $E: M_n \rightarrow D_n$ which acts by erasing off-diagonal entries. Theorem 1.1 concerns the preimage of $(1/2)\text{id}$ under the restriction of E to projections. It is natural to ask if preimages of other matrices are connected as well. We do not have a complete answer, see however Theorem 4.1. The image of the set of projections under E was the subject of recent papers by Kadison [11, 12].

Theorem 8.1 provides a partial extension of Theorem 1.1 to infinite-dimensional spaces, where the notion of connectedness is understood in the sense of norm topology on the space of bounded operators.

Our second main result is a non-self-adjoint version of Theorem 1.1 which applies to idempotent matrices with an arbitrary fixed diagonal.

Theorem 1.2. *For every d in $D_n(\mathbb{C})$, the set of idempotents q in $M_n(\mathbb{C})$ such that $E(q) = d$ is pathwise connected.*

projection has trivial relative commutant form a path-connected set.

The paper concludes with Section 8, where some of our results are extended to operators in separable infinite-dimensional Hilbert spaces. Whether full analogues of Theorem 1.1 and 1.2 hold in infinite dimensions remains open.

2. PRELIMINARIES

2.1. Projections as 2×2 matrices. The content of this section is well-known folklore. It is essentially contained in [8, Theorem 2]. We include this discussion for the convenience of the reader, since it is the basis for our proof of Theorem 1.1.

When a Hilbert space comes as an orthogonal direct sum of two Hilbert spaces, say $\mathcal{H} = K \oplus L$, projections p of $B(\mathcal{H})$ can be identified with those 2×2 matrices

$$p = \begin{pmatrix} a & b \\ b^* & d \end{pmatrix}$$

where a, b, d are operators in $B(K), B(L, K), B(L)$ respectively, such that

$$(i) \quad 0 \leq a \leq \text{id} \text{ and } 0 \leq d \leq \text{id};$$

The idempotent half of that literal question has a negative answer, even for a fixed diagonal which is not identically zero.

Idea of the proof

The key is to exploit a phenomenon unavailable in finite dimensions: an infinite-rank idempotent can have zero diagonal. We build one explicitly from infinitely many copies of a rank-one 2×2 oblique projection. A countable regrouping places two coordinates carrying diagonal weight -1 beside one carrying weight 2 ; an orthogonal 3×3 mixing gives every new basis vector weights $2/3$ and $1/3$, so its diagonal value is $-(2/3) + 2(1/3) = 0$.

After adjoining a one-dimensional identity summand, this idempotent and the rank-one coordinate projection have the same nonzero diagonal $(1, 0, 0, \dots)$. They cannot be joined by a norm-continuous path of idempotents because rank is locally constant in the idempotent set.

Counterexample

Throughout, $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$, and diagonals are taken relative to the displayed orthonormal basis.

Theorem 1. *There exist a separable infinite-dimensional Hilbert space H over $\mathbb{F}$, an orthonormal basis $(e_j)_{j \geq 0}$, and the nonzero diagonal $d = (1, 0, 0, \dots)$ such that*

$$\mathcal{I}_d = \{T \in B(H) : T^2 = T, \langle Te_j, e_j \rangle = d_j \text{ for all } j\}$$

is not path-connected in the operator norm. More precisely, $\mathcal{I}_d$ contains a rank-one idempotent and an infinite-rank idempotent, which lie in different norm-path components.

Proof. Let $K = \ell_2(\mathbb{N}) \oplus \ell_2(\mathbb{N})$, with its usual orthonormal vectors u_i, v_i for $i \in \mathbb{N} = \{0, 1, 2, \dots\}$. Define $R \in B(K)$ blockwise by

$$Ru_i = -u_i - 2v_i, \quad Rv_i = u_i + 2v_i. \tag{1}$$

On $\text{span}\{u_i, v_i\}$ this is the matrix

$$A = \begin{pmatrix} -1 & 1 \\ -2 & 2 \end{pmatrix}, \quad A^2 = A.$$

Thus $R^2 = R$. Every block has one-dimensional range $\mathbb{F}(u_i + 2v_i)$, so R has infinite rank.

We now exhibit an orthonormal basis in which R has zero diagonal. Define a bijection $\phi : \mathbb{N} \times \{0, 1\} \rightarrow \mathbb{N}$ by

$$\begin{array}{c|cccc} (m, j) & (0, 0) & (0, 1) & (1, 0) & (1, 1) \\ \hline \phi(m, j) & 1 & 2 & 3 & 0, \end{array}$$

and, for $m \geq 2$, by

$$\phi(m, 0) = 2m + 1, \quad \phi(m, 1) = 2m.$$

For every m and j one has $\phi(m, j) \neq m$. The three-dimensional spaces

$$G_m = \text{span}\{u_{\phi(m,0)}, u_{\phi(m,1)}, v_m\}$$

are mutually orthogonal and their orthogonal direct sum is K .

In each G_m , let $w_{m,0}, w_{m,1}, w_{m,2}$ be the vectors whose coordinate columns relative to the displayed ordered spanning set are the columns of

$$O = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix}. \quad (2)$$

The matrix O is real orthogonal. Hence all the $w_{m,r}$ form an orthonormal basis of K , over either scalar field. Each column of O has total squared weight $2/3$ in its first two coordinates and squared weight $1/3$ in its last coordinate.

Write one such vector as

$$w = \alpha u_{\phi(m,0)} + \beta u_{\phi(m,1)} + \gamma v_m.$$

The two inequalities $\phi(m, j) \neq m$ ensure that none of the cross terms in (1) contributes to $\langle Rw, w \rangle$. Consequently

$$\langle Rw, w \rangle = -|\alpha|^2 - |\beta|^2 + 2|\gamma|^2 = -\frac{2}{3} + 2\frac{1}{3} = 0.$$

Thus R is an infinite-rank idempotent with zero diagonal in the basis $(w_{m,r})$.

Finally let $H = \mathbb{F}z \oplus K$, where z is a unit vector, and order the basis as

$$e_0 = z, \quad e_{3m+r+1} = w_{m,r} \quad (m \geq 0, 0 \leq r \leq 2).$$

Consider

$$P = I_{\mathbb{F}z} \oplus 0_K, \quad Q = I_{\mathbb{F}z} \oplus R.$$

Both are idempotents and both have diagonal $(1, 0, 0, \dots)$. But $\dim \text{Ran } P = 1$, whereas $\dim \text{Ran } Q = \aleph_0$.

It remains to show that this prevents a norm-continuous idempotent path. If S, T are idempotents with $\|S - T\| < 1$, then $T : \text{Ran } S \rightarrow \text{Ran } T$ is injective: for $x \in \text{Ran } S$ with $Tx = 0$,

$$\|x\| = \|Sx\| = \|(S - T)x\| < \|x\|,$$

unless $x = 0$. Interchanging S and T gives injections in both directions, so their Hilbert-space ranks agree. Any norm-continuous path on $[0, 1]$ is uniformly continuous and can be subdivided into finitely many consecutive steps of norm less than one. Rank is therefore constant along every such idempotent path. Since P and Q have different ranks, they lie in different path components of $\mathcal{I}_d$. $\square$

Verification

The proof uses only the block identity $A^2 = A$, orthogonality of O , the explicit bijection ϕ , and the local rank lemma. The included script `code/verify_construction.py` checks the block identity and rank, the orthogonality and weight cancellation in (2), bijectivity and absence of forbidden pairings through 2,000 groups, and 6,000 initial diagonal values. Run it from the repository root with

```
conda run --no-capture-output -n sandbox python \  
  runs/fa_banach_001/solutions/counterexamples/  
  0906.0139_infinite_fixed_diagonal_idempotents_disconnected/  
  code/verify_construction.py
```

where the three path lines are concatenated. These checks are corroborative; the displayed identities constitute the proof.

Scope and novelty check

This counterexample settles only the idempotent half of the source’s final infinite-dimensional question. It does not decide whether all projections with constant diagonal $1/2$ are norm-path-connected. Eight focused upgrade attempts tested block-index, compact-perturbation, paving, and finite approximation routes; no global invariant or uniform homotopy was obtained for that projection fiber.

The finite frame-homotopy problem is no longer open: Cahill, Mixon, and Strawn explicitly cite the source and completely resolve it [2]. Needham and Shonkwiler’s prescribed-frame theorem also implies, after the usual Gramian quotient, that every nonempty finite-dimensional complex projection fiber with arbitrary prescribed diagonal is path-connected [3].

The bounded search was run on 17 August 2026. It covered the run indexes and primary-source/arXiv queries using the source identifier and exact variants of “fixed diagonal idempotents path connected,” “zero-diagonal idempotents,” and the source’s question. It found the known zero-diagonal existence theory—the source itself notes such examples, and Loreaux–Weiss later prove much stronger diagonal-realization results [4]—but no explicit statement of the nonzero-fiber rank-component counterexample above. Novelty confidence is therefore limited: the construction and topological conclusion here are self-contained, but the underlying zero-diagonal phenomenon is known.

Human-review recommendation

Check (i) that ϕ is a bijection and the G_m exhaust K , (ii) that the forbidden equalities $\phi(m, j) = m$ are exactly what eliminates the cross terms, and (iii) that the intended infinite analogue of Theorem 1.2 does not silently restrict to one fixed rank/nullity stratum. Under the literal formulation recorded in the source introduction, the counterexample is complete.

References

- [1] J. Giol, L. V. Kovalev, D. Larson, N. Nguyen, and J. E. Tener, *Projections and idempotents with fixed diagonal and the homotopy problem for unit tight frames*, Operators and Matrices 5 (2011), 139–155; arXiv:0906.0139.

- [2] J. Cahill, D. G. Mixon, and N. Strawn, *Connectivity and irreducibility of algebraic varieties of finite unit norm tight frames*, SIAM J. Appl. Algebra Geom. 1 (2017), 38–72; arXiv:1311.4748.
- [3] T. Needham and C. Shonkwiler, *Symplectic geometry and connectivity of spaces of frames*, Adv. Comput. Math. 47 (2021), article 5; arXiv:1804.05899.
- [4] J. Loreaux and G. Weiss, *Diagonality and idempotents with applications to problems in operator theory and frame theory*, J. Operator Theory 75 (2016), 91–118; arXiv:1410.7441.